



DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

Review : Major Project Phase-II (IS2281)

Design and Implementation of a Smart Health Monitoring and Early Warning System for Water-Borne Diseases

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ABSTRACT

Traditional rural health systems face delays in identifying water-borne disease outbreaks due to the lack of integration between citizen-reported symptoms and environmental data. This system proposes a Smart Health Monitoring and Early Warning System based on a human-in-the-loop framework, where citizen inputs are validated by ASHA workers to ensure data reliability. A multidimensional dataset of 3000 records with 15 parameters is processed using a stacking ensemble model combining Support Vector Machine (SVM), Random Forest, and Gradient Boosting. The model performs risk classification into Low, Medium, and High categories, achieving 89.6% accuracy with strong recall for high-risk cases. The system follows pipeline for data collection, validation, preprocessing, and prediction, enabling real-time alert generation for health officials. This approach improves prediction robustness and reduces response time for outbreak control.



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2. Literature Survey
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PROBLEM STATEMENT

Water-borne diseases such as Cholera, Typhoid, and Dysentery remain a major concern in developing regions due to poor water quality, inadequate sanitation, and lack of awareness. Existing healthcare systems primarily rely on hospital-based and laboratory reporting, which leads to delayed identification of disease outbreaks. The absence of real-time monitoring mechanisms and early-stage detection results in rapid disease spread, increased healthcare burden, and higher mortality rates in vulnerable communities. Current systems lack a coordinated framework for collecting and analyzing data from multiple sources, such as community-level symptoms and environmental conditions. This limitation reduces the effectiveness of timely intervention and highlights the need for an improved surveillance approach.



LITERATURE SURVEY

SL. No.	Author & Year of Publication	Title of Article	Methods Used
1	Getahun et al.,2024 [1]	IoT based automated disease prediction system	LSTM + time-series Augmentation + IoT Series
2	Zhang et al.,2023 [2]	Real-time water quality monitoring using low-cost IoT sensors	IoT sensor based real monitoring
3	Nguyen et al., 2024 [3]	Predictive modeling of water-borne disease outbreaks	ML using environmental & epidemiological data
4	Patel & Singh, 2023 [4]	Digital health transformation in rural India	Analysis of telemedicine & mHealth systems
5	Ahmed & Thompson, 2024 [5]	Sustainable IoT-based health monitoring systems	LoRaWAN + solar-powered IoT networks



LITERATURE SURVEY

SL. No.	Author & Year of Publication	Title of Article	Methods Used
6	GSMA, 2024 [6]	The Mobile Economy 2024	Global mobile usage analysis
7	Kumar & Verma, 2021 [7]	Smart water quality monitoring system	Threshold analysis + sensor based data
8	Essamlali et al., 2024 [8]	ML and IoT for water quality monitoring	Review of IoT + ML integration
9	Vasudevan & Baskaran, 2023 [9]	Hybrid ML and IoT water monitoring system	Embedded IoT + Hybrid ML
10	Kumar et al., 2024 [10]	Artificial intelligence in rural health surveillance: Opportunities and challenges	Review of AI applications in health surveillance



RESEARCH GAP AND INNOVATION

RESEARCH GAPS	System USP
● Absence of Human-in-the-Loop Validation: No mechanism for ground-level verification of symptoms and environmental conditions.	● Human-in-the-Loop Validation: Combines citizen reporting with ASHA worker verification to ensure reliable data, unlike purely sensor-based systems.
● Fragmented & Limited Data Integration: Most systems use either water quality or health data, lacking a unified real-time pipeline.	● Integrated Multi-Source Data Pipeline: Combines health symptoms, water quality, environmental data for more accurate prediction.
● Passive Surveillance & No Community Input: Reliance on hospital records ignores early symptom reporting from citizens.	● Active Community-Driven Surveillance: Enables real-time citizen symptom reporting, transforming passive hospital-based systems into a proactive early detection mechanism.
● Limited Decision Support & Actionability: Systems provide predictions but lack meaningful insights and alert mechanisms for effective decision-making by health officials.	● Actionable Decision Support System: Generates validated risk alerts (Low/Medium/High) and directly notifies health officials, enabling timely intervention and resource allocation



INTRODUCTION

Water-borne diseases such as Cholera, Typhoid, and Dysentery remain a major public health challenge in rural regions due to poor sanitation and unsafe water. Traditional systems rely on passive hospital based reporting, leading to delayed outbreak detection. Although IoT and machine learning enable automated monitoring, many solutions lack reliable validation and integrated multi-source data processing. This system proposes a Smart Health Monitoring and Early Warning System based on a layered architecture and a stacking ensemble model (SVM, Random Forest, Gradient Boosting) to classify outbreak risk and provide effective **decision support** for public health management..

SYSTEM DESIGN

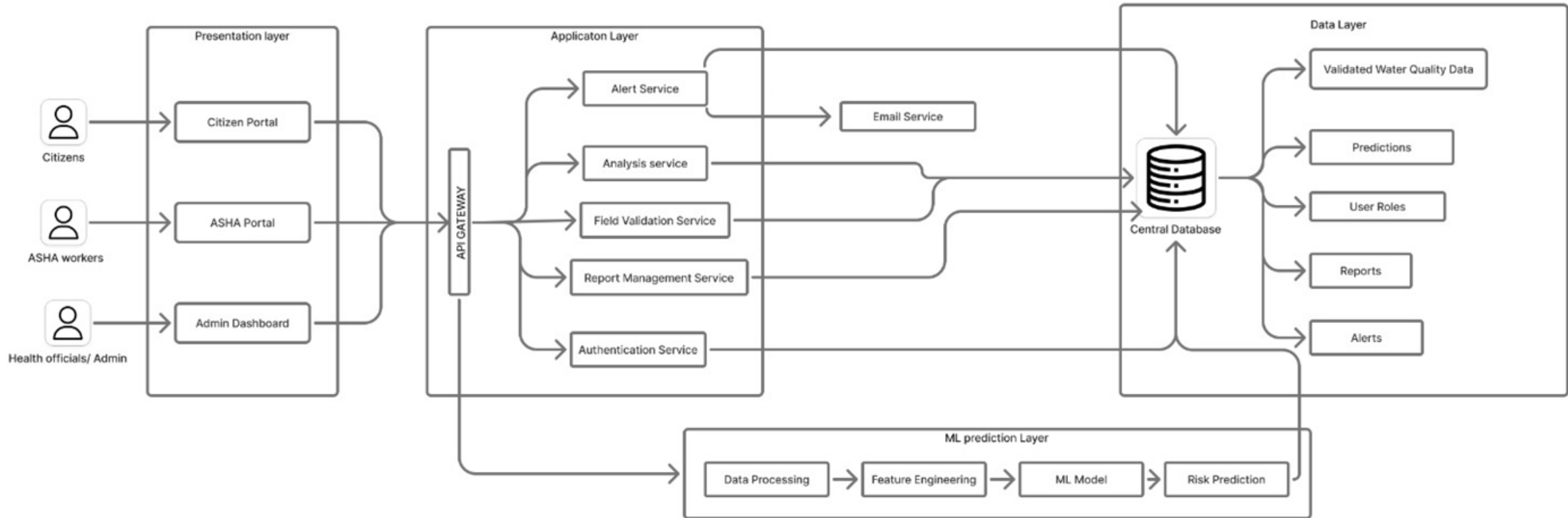


Fig. - 1: System Architecture

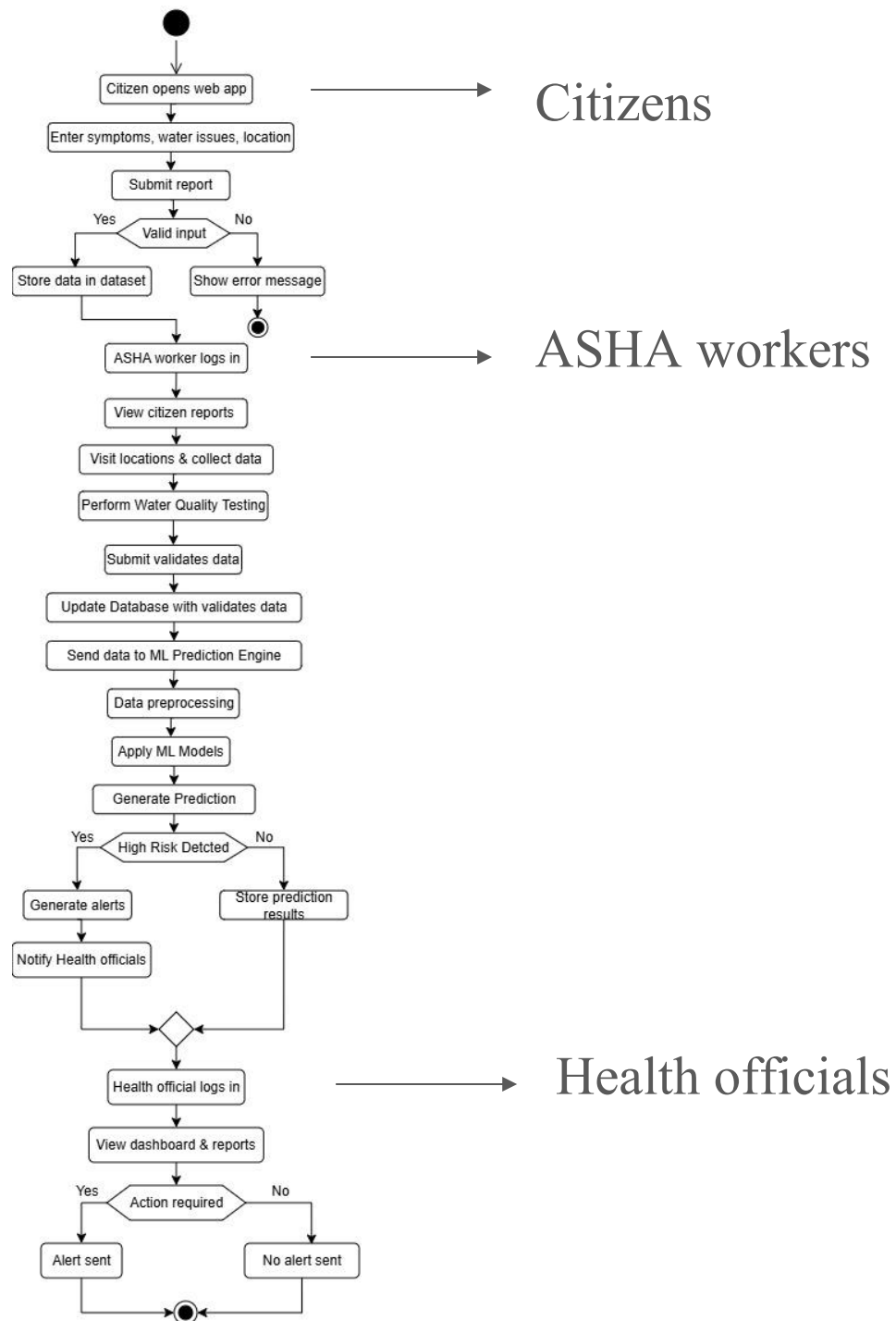


Fig. - 2: Activity diagram

SEQUENCE DIAGRAM

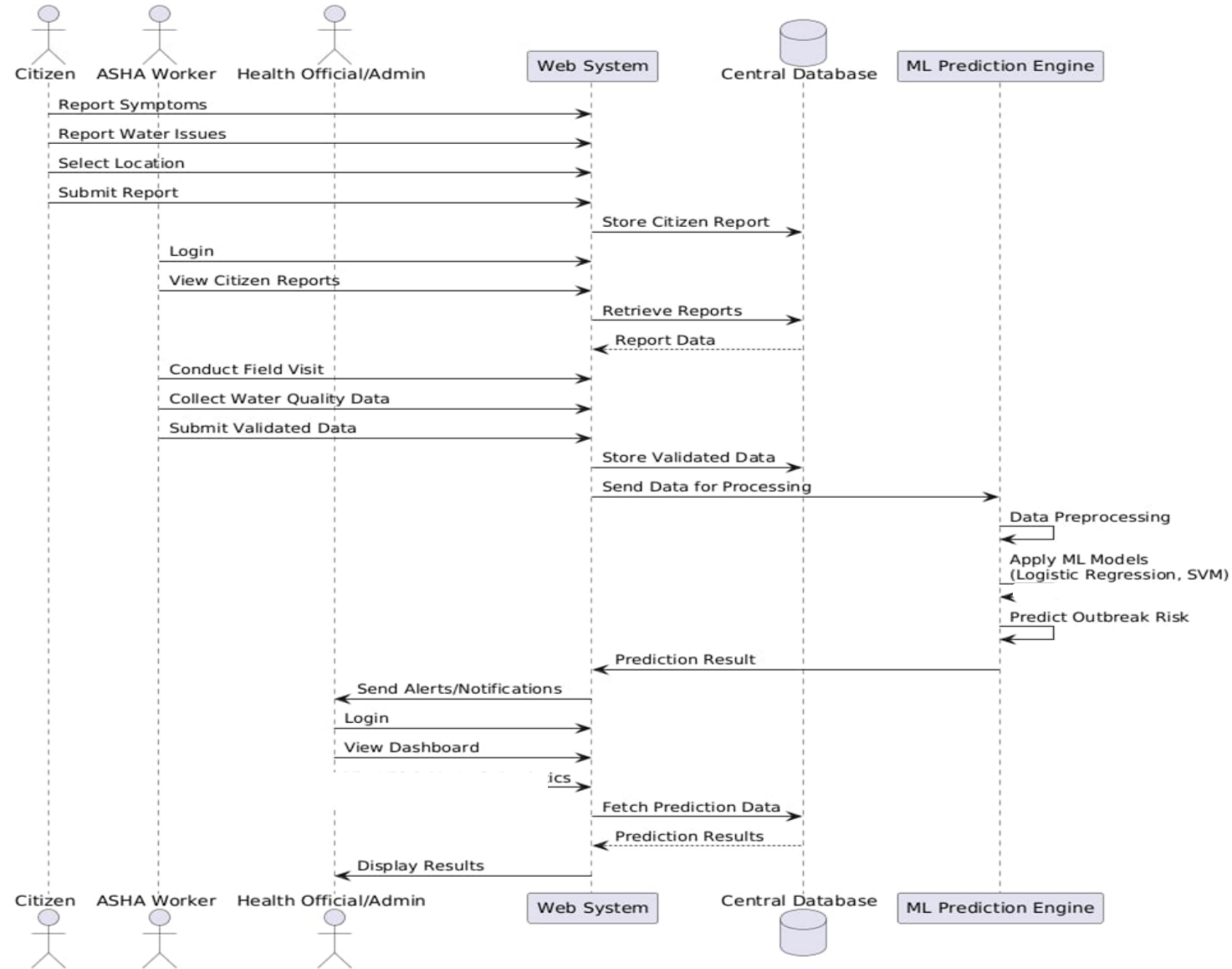


Fig. - 3: Sequence Diagram



SYSTEM METHODOLOGY OVERVIEW

The system follows a layered architecture for disease surveillance and prediction.

1. Presentation Layer

- Provides the user interface for Citizens, ASHA Workers, and Health Officials.
- Allows users to submit reports, view data, and access dashboards.
- Ensures easy and interactive communication with the system.

2. Application Layer

- Handles the main business logic and request processing.
- Manages authentication, report validation, alerts, and dashboard functions.
- Connects the user interface with the database and ML engine.



SYSTEM METHODOLOGY OVERVIEW

3. Data Layer

- Stores citizen reports, validated water-quality data, and prediction results.
- Manages secure retrieval and updating of system records.
- Acts as the central repository for all application data.

4. ML Prediction Layer

- Preprocesses the collected data for analysis.
- Applies machine learning models to detect outbreak risk.
- Generates prediction results and risk alerts for officials.



SYSTEM METHODOLOGY OVERVIEW

5. Prediction using Ensemble Model

- Processed data is passed to:
 - Support Vector Machine (SVM)
 - Random Forest
 - Gradient Boosting
- Combined using **stacking ensemble technique**

6. Alert & Notification

- High and Medium risk cases trigger **alert service**
- Notifications sent via **email service**
- Enables **timely decision-making by health officials**



MACHINE LEARNING & PREDICTION PROCESS

1. Data collection

- Dataset: **3000 records, 15 features**
- Includes:
 - Water Quality (pH, turbidity, bacterial count)
 - Environmental factors (rainfall, temperature)
 - Health indicators (fever, diarrhea)

2. Data Preprocessing

- Handling missing values and noise
- Feature scaling using **StandardScaler**
- Data normalization for consistent model learning

3. Feature Engineering

- Creation of domain-specific features:
 - **Water Quality Index (WQI)**
 - **Health Vulnerability Score**
 - Interaction features (e.g., bacterial count $\times$ water scarcity)
- Improves model accuracy and pattern detection

6. Alert & Decision Support

- Predictions stored in database
- Medium/High risk triggers **automated alerts**
- Enables real-time intervention by health officials

5. Prediction Process

- Input data $\rightarrow$ scaled using saved scaler
- Passed to trained ensemble model
- Output classified into:
 - Low Risk
 - Medium Risk
 - High Risk

4. Model Training & Ensemble Learning

- Base Models:
 - Support Vector Machine (SVM)
 - Random Forest
 - Gradient Boosting
- Advanced Techniques:
 - Hyperparameter tuning using GridSearchCV
 - Stacking Ensemble Model



IMPLEMENTATION

The system implements a pipeline integrated with a Flask-based backend for real-time prediction and alerting

1. Model Development Pipeline

- **Input Schema:** 15 features (water quality, environmental, health indicators)
- **Target Classes:** Low, Medium, High
- **Preprocessing & Feature Engineering:**
 - Missing value imputation
 - Feature scaling & normalization
 - Derived features:
 - Water Quality Index (WQI)
 - Health Vulnerability Score
 - Interaction features



IMPLEMENTATION

- **Model Training & Optimization:**

- Algorithms: SVM, Random Forest, Gradient Boosting
- Hyperparameter tuning using **GridSearchCV**
- Ensemble methods: **StackingClassifier, VotingClassifier**

- **Model Serialization:**

- Final model and scaler stored using **Joblib**
- Enables direct use in deployment via `transform()` and `predict()`

2. Inference & Prediction Pipeline

- Model and scaler loaded at service startup
- Input received via **REST API (/api/predict)**



IMPLEMENTATION

- Steps:
 - Input validation and formatting
 - Feature transformation and scaling
 - Model inference → class prediction
 - Mapping to risk levels into Low / Medium / High

3. Data Persistence

- Prediction results stored in **SQLite database**
- Includes:
 - Input features
 - Predicted risk level
 - Timestamp and location



IMPLEMENTATION

4. Alert & Notification System

- Rule-based alert mechanism:
 - High Risk → Immediate alert
 - Medium Risk → Monitoring alert
- Low Risk → No alert
- Email integration using **SMTP (smtplib)**:
 - Secure connection via **STARTTLS**
 - Alerts sent to health officials
- Fault-tolerant design:
 - Alert failure does not affect prediction process



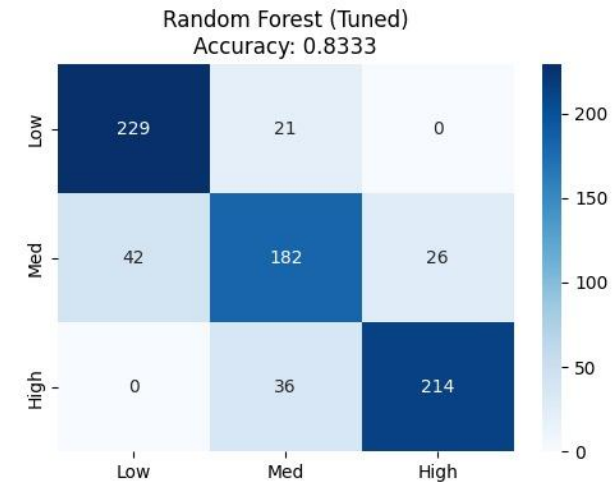
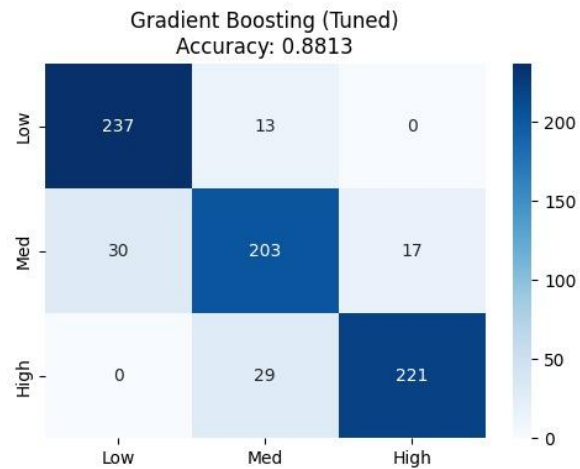
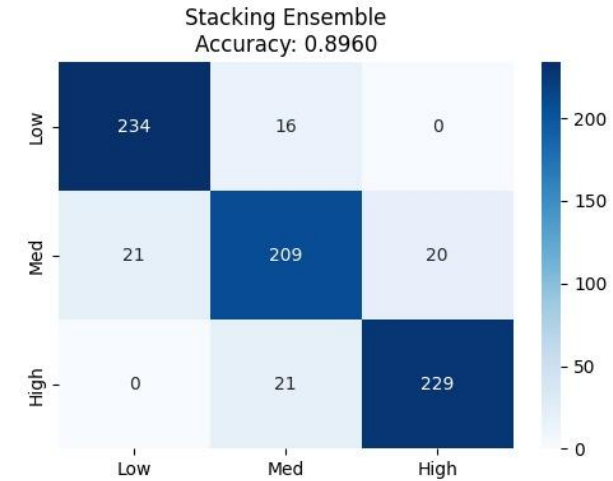
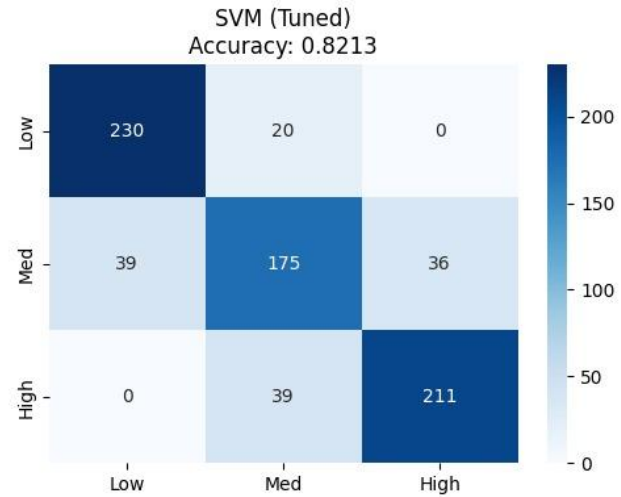
IMPLEMENTATION

5. End-to-End Execution Flow

- **ASHA Input → Validation → Feature Transformation → ML Prediction → Risk Classification → Database Storage → Alert Generation**



IMPLEMENTATION



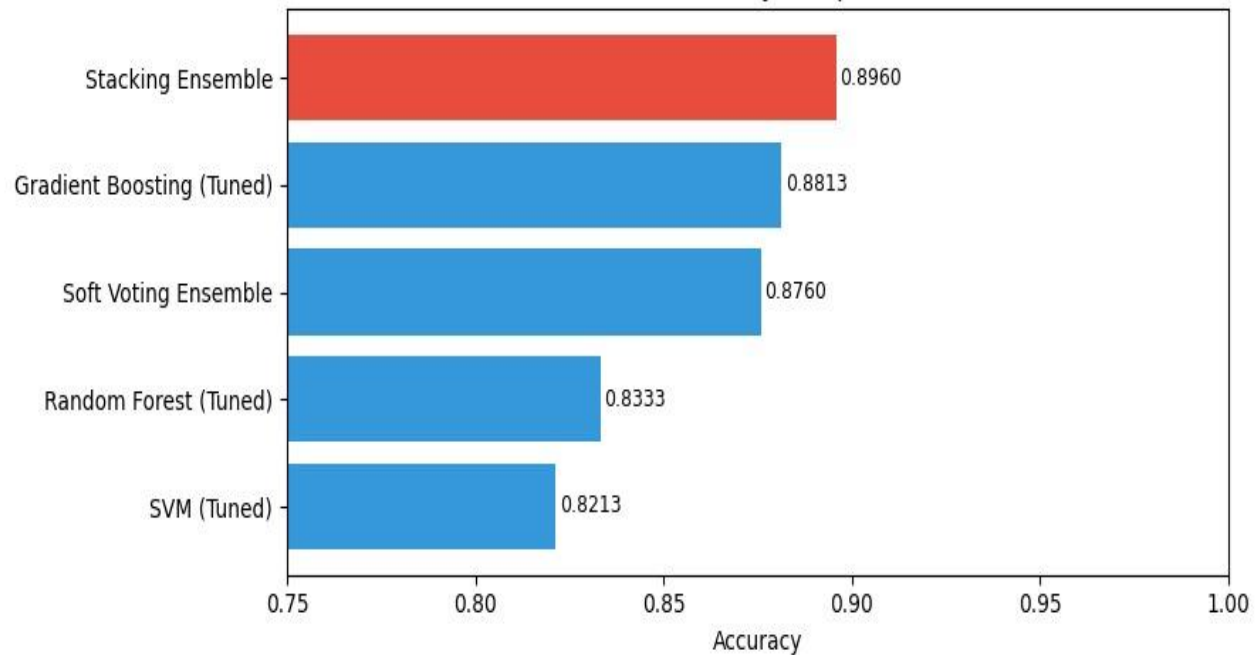
RESULTS

“Overall Accuracy: 89.6%”

Model Performance & Metrics

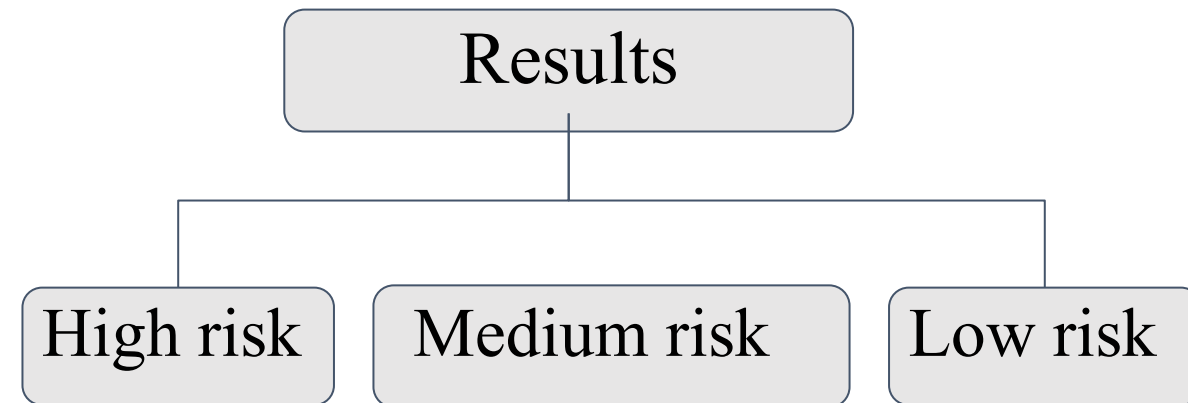
Stacking Ensemble Outperformed

Model Accuracy Comparison



Classification results

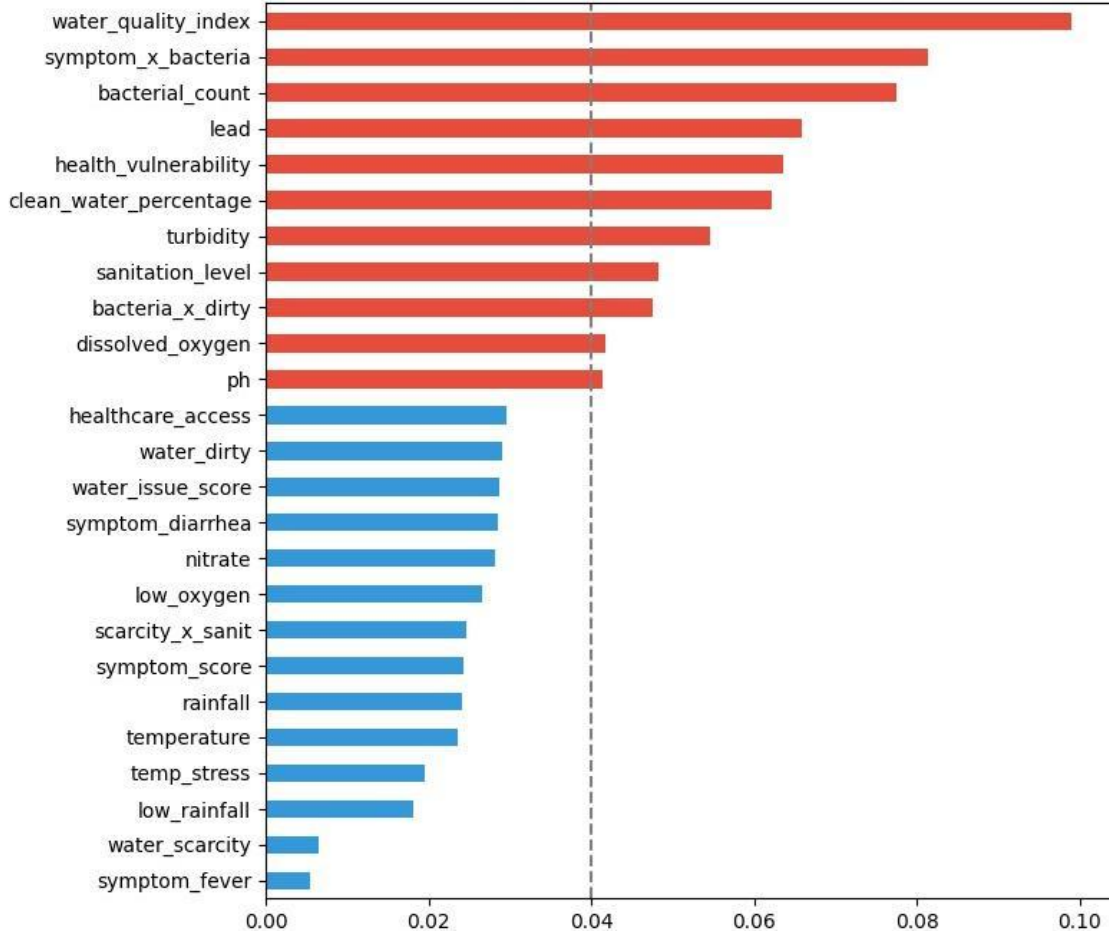
Strong Discrimination Ability



RESULTS

Feature Impact & Importance

Improved prediction accuracy



System & Practical Impact

- Real-time data collection & alerts.
- Better citizen-ASHA worker coordination.
- Reduces delay in identifying outbreak zones.
- Enables early intervention & prevention.
- Data-driven decision-making for authorities.

RESULTS

ALERT NOTIFICATIONS

HIGH RISK ALERT - Water-Borne Disease Outbreak Risk in Bengaluru Inbox ☆

1 1ds22is002@... Yesterday 😊 ↩ ⋮
to me ▾

HIGH RISK ALERT - IMMEDIATE ACTION REQUIRED

=====

WATER-BORNE DISEASE SURVEILLANCE SYSTEM

=====

A HIGH RISK prediction has been recorded in your area!

Location Details:

Area: Bengaluru
Reported by: asha1 (ASHA Worker)
Time: 2026-04-11 18:10:18

Water Quality Parameters:

- Temperature: 32°C
- pH Level: 7
- Turbidity: 444 NTU
- Bacterial Count: 4 CFU/mL
- Dissolved Oxygen: 4 mg/L
- Nitrate Level: 4 mg/L
- Lead Level: 4 mg/L

Infrastructure Status:

- Clean Water Availability: 4%
- Sanitation Level: 4%
- Healthcare Access: 4%

Reported Symptoms:

- Diarrhea: No
- Fever: Yes

Water Issues:

- Dirty Water: No
- Water Scarcity: No

Environmental Factors:

- Rainfall: 222 mm

RISK ASSESSMENT: HIGH

RECOMMENDED ACTIONS:

1. IMMEDIATE: Alert local health authorities
2. Deploy medical team for health screening
3. Distribute chlorine tablets/water purifiers
4. Issue public health advisory in affected area
5. Conduct additional water quality tests
6. Set up temporary health camps
7. Launch awareness campaign on hygiene

Response Timeline: Action required within 24 hours

This is an automated alert from Water-Borne Disease Surveillance System.
Please take immediate action to prevent potential outbreak.

—
System Administrator; Contact for any issues

↩ Reply ↗ Forward 😊

MEDIUM RISK Alert - Monitor Situation in Bengaluru Inbox ☆

1 1ds22is002@dsce.edu.in 11:09 pm
MEDIUM RISK ALERT - Monitor Required
=====

1 1ds22is002@ds... 11:15 pm 😊 ↩ ⋮
to me ▾

MEDIUM RISK ALERT - Monitor Required

=====

Location: Bengaluru
ASHA Worker: asha1
Risk Level: MEDIUM
Time: 2026-04-12 23:15:32

Key Parameters:

- Bacterial Count: 49 CFU/mL
- Clean Water: 81%

[Show quoted text](#)

↩ Reply ↗ Forward 😊



SCOPE OF FUTURE WORK

- Integration of IoT-based water quality sensors for real-time monitoring
- Development of a mobile application for easier citizen and ASHA access
- Deployment on cloud platforms for scalability and real-time access
- Integration with government health systems for centralized monitoring
- **Future Vision:**
Transform the system into a smart, real-time public health surveillance platform for large-scale deployment.



REFERENCES

- [1] M. Getahun, V. Kulkarni, S. Mondal, G. S. P. Ghantasala, A. Al-Rasheed, and B. O. Soufiene, "IoT-based automated system for water-related disease prediction," *Scientific Reports*, vol. 14, p. 29483, 2024.
- [2] I. Essamlali, H. Nhaila, and M. El Khaili, "Advances in machine learning and IoT for water quality monitoring: A comprehensive review," *Heliyon*, vol. 10, no. 6, p. e27920, 2024.
- [3] A. Kumar and S. Verma, "Smart water quality monitoring system for early detection of water-borne diseases," *Environmental Monitoring Systems*, vol. 9, no. 1, pp. 45-62, 2021.
- [4] M. Rahman, "Predictive modeling of water-borne disease outbreaks using environmental and epidemiological data," *Environmental Health Perspectives*, vol. 132, no. 2, p. 027001, 2024.
- [5] K. Patel and A. Singh, "Digital health transformation in rural India: Lessons from the pandemic," *Journal of Rural Health Technology*, vol. 9, no. 4, pp. 312-328, 2023.
- [6] R. Ahmed and L. Thompson, "Sustainable deployment of IoT-based health monitoring systems in resource-limited rural settings," *Journal of Global Health Informatics*, vol. 12, no. 3, pp. 245-261, 2024.
- [7] GSMA Intelligence, "The Mobile Economy 2024," *GSMA Intelligence Report*, 2024.
- [8] R. Kumar and S. Verma, "Smart water quality monitoring system using threshold analysis and sensor-based data," 2021.
- [9] A. Essamlali et al., "Machine learning and IoT for water quality monitoring: A review of IoT and ML integration," 2024.
- [10] S. Vasudevan and B. Baskaran, "Hybrid machine learning and IoT-based water monitoring system using embedded IoT and hybrid ML," 2023.



THANK YOU